

Spatial Risk Analysis

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We want to investigate risks in the context of climate extremes and their spatial distribution. For this, we build up on the notion of spatial risk measures developed by Koch 2017. We briefly recapitulate the concepts: Assume that we have a max-stable process $\{Z(\mathbf{x})\}_{\mathbf{x} \in \mathbb{R}^2}$ that has been fitted to spatial data of climate extremes (that have been transformed to Fréchet margins before). Assume further that a damage function $D : \mathbb{R} \rightarrow \mathbb{R}$ and an exposure process $\{E(\mathbf{x})\}_{\mathbf{x} \in \mathbb{R}^2}$ are given (both deterministic). The damage function is used to map the extreme values to the associated damage (e.g. financial damage or loss of lives), and the exposure process describes for each point in space how vulnerable it is to the effects of climate extremes. Taking these together, we obtain a cost process which describes the (random) costs caused by extremes at any location:

$$\{C(\mathbf{x})\}_{\mathbf{x} \in \mathbb{R}^2} = \{E(\mathbf{x})D(Z(\mathbf{x}))\}_{\mathbf{x} \in \mathbb{R}^2}.$$

In classical risk theory, the risk that is associated with the cost (which is a univariate random variable in classical theory) is determined by applying a certain risk measure to it. This measure could be the expectation, the variance, or a high quantile ("value at risk"), to name a few. If the risk is to be determined over an area, the individual risks at each point of it are usually simply averaged. But this does not take into account spatial dependency and the fact that extremes can have a tendency to occur simultaneously at different locations, increasing the overall risk. The solution that Koch 2017 propose is basically to switch the order of averaging and applying the risk measure: They define a spatially averaged cost (or spatially averaged loss) as

$$L(A) = \frac{1}{|A|} \int_A C(\mathbf{x}) \, d\mathbf{x},$$

for A being a compact subset of $\mathbb{R}^2$, and then apply a risk measure to $L(A)$, thus taking into account possible spatial dependencies.

[Part that is not formally written yet:] To test the application of these ideas I used historical global temperature data from the MPI-ESM model. One possible idea is to investigate deaths and/or health-related issues caused by extreme temperatures. We need a cost function and an exposure measure for that. As a first rough approximation, I defined the cost function $C(x)$ as 0 for $x < 30$, 1 for $x > 40$ and linear in between. Later, this must of course be replaced by something that's somehow justified, I'm trying to find something in the literature (like mortality rates depending on temperature, I'm sure something can be found). As exposure process I use the total population within the grid cells of the climate model (also here, something more sophisticated can be used, taking into account not only population, but also e.g. living conditions or healthcare standards). As risk measure I used the expected shortfall at 5% level. The investigated areas A include for each grid point the grid point itself and all points within 3 grid points distance. As comparison, the expected shortfall for each grid point (without taking into account spatial dependencies) is shown in Fig. 2.

Koch 2017 provides explicit formulas for the risk associated with various risk measures and different kinds of max-stable processes. The problem is that they require both spatial stationarity of the dependence structure and marginal distributions that don't depend on the location. Especially the last requirement seems realistic only for very small areas of investigation. For the investigations now, those formulas were not useable therefore, and I instead used Monte-Carlo to calculate the risks, using local estimates of the marginal parameters and for the dependency structure at each grid point (assuming that they are approximately constant on the investigated area A). The resulting risks are shown in the Figure below, and we see risks being connected to high temperature and high population density. Of course, the actual values are not at all correct or informative yet, for that we'd need accurate exposure and damage functions, the figure serves more or less only as a proof of concept for now.

An idea would be to apply both classical risk measures and the new spatial risk measures to some data and compare the results, and how the inclusion of spatial dependencies influences risk assessments. I previously thought about combining risk measures with clusterings, but I'm not so sure about that anymore, because the results especially at the cluster boundaries would be very artificial (if we assume constant dependency structures within each clusters, then there necessarily must have abrupt shifts in the structures when moving from one cluster to another, and that's not quite plausible). So I think a non-parametric approach using local estimates for each grid point investigated is more promising.

It is also possible to assume that the dependency structure is constant on a certain region and that a simple parametric model can be used for the marginal parameters (i.e. linear regression using latitude, longitude and elevation as covariates) and maybe to extend the formulas given by Koch 2017 to that case.

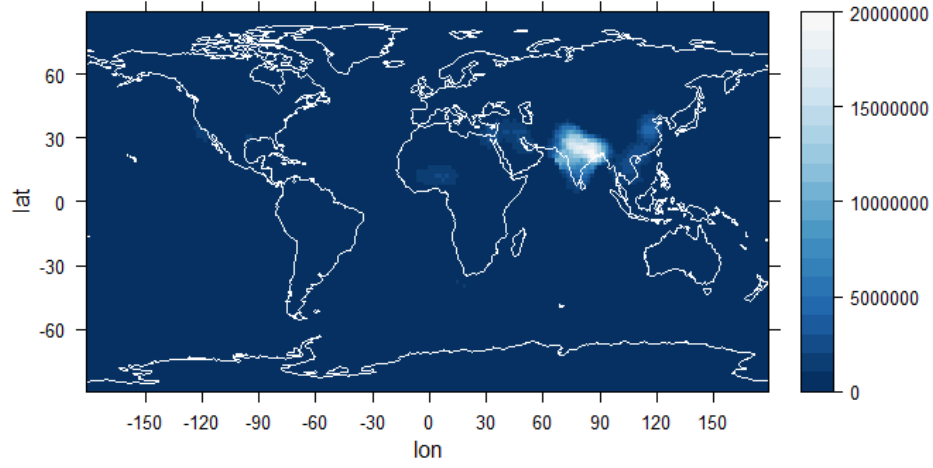


Figure 1: The temperature-related expected shortfall at 5% level, using for each grid point a spatially averaged risk based on all grid points with a distance of 3 or less to the point.

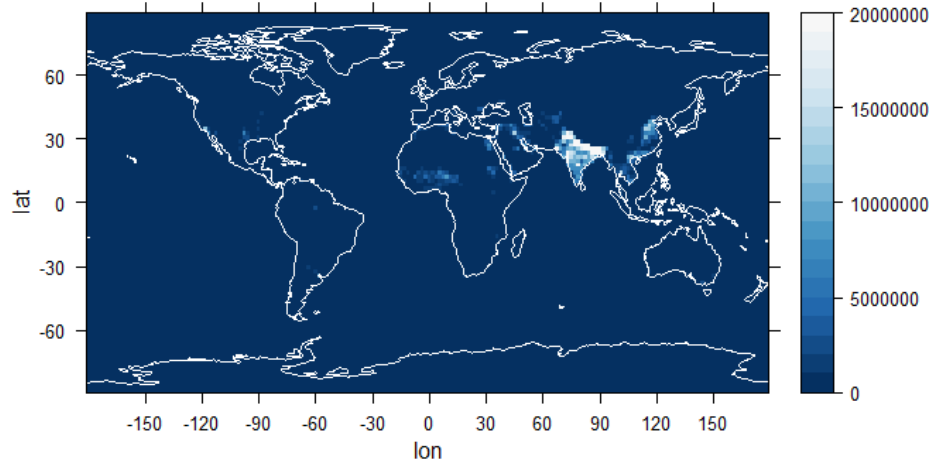


Figure 2: The temperature-related expected shortfall at 5% level, calculated for each grid point without taking into account spatial dependencies.

References

Koch, Erwin (2017). “Spatial risk measures and applications to max-stable processes”. In: *Extremes* 20, pp. 635–670. DOI: 10.1007/s10687-016-0274-0.